

Тест Blockchain

```
D:\Codes\technology_rasp_reestr\1 lab\merkle-tree>npx mocha test
```

Blockchain

```
✓ should be considered valid
tampering with a previousHash
✓ should not be considered valid
tampering with data
✓ should not be considered valid
```

Merkle proof verification

a given merkle tree

untampered proofs

```
✓ should verify the proof for leaf index 0
✓ should verify the proof for leaf index 1
✓ should verify the proof for leaf index 2
✓ should verify the proof for leaf index 3
✓ should verify the proof for leaf index 4
✓ should verify the proof for leaf index 5
✓ should verify the proof for leaf index 6
✓ should verify the proof for leaf index 7
✓ should verify the proof for leaf index 8
✓ should verify the proof for leaf index 9
✓ should verify the proof for leaf index 10
```

tampered proofs

verifying a different node with a proof

```
✓ should not verify the proof
```

verifying a different root

```
✓ should not verify the proof
```

flipping a nodes position

```
✓ should not verify the proof
```

editing a hash

```
✓ should not verify the proof
```

Тест Transation

Transation

constructor

```
✓ should set the owner
✓ should set the amount
✓ should set the address
✓ should set spent to false
```

spend

```
✓ should set spent to true
✓ should not spent twice
```

hash

```
✓ should have valid hash
```

Тест Binary search tree

Binary search tree

- ✓ should have a null root
- after adding 5
 - ✓ should have a root
 - ✓ should find 5
 - ✓ should not find 4
- after adding 3
 - ✓ should have add a left to the root
 - ✓ should find 3
 - ✓ should not find 4
- after adding 4
 - ✓ should have add to the left node
 - ✓ should find 4
- after adding 7
 - ✓ should have add a right to the root
 - ✓ should find 7

Tect Trie

Trie

with a single word

- ✓ should connect the root to the first letter
- ✓ should connect the root to the second letter
- ✓ should connect the root to the third letter

with three words

for helipad

- ✓ should connect to the final letter

for hello

- ✓ should connect to the final letter

for hermit

- ✓ should connect to the final letter

42 passing (18ms)

D:\Codes\technology_rasp_reestr\1 lab\merkle-tree>